

ZigBee Applications

Purpose	Application areas		
	Home automation	Building automation	Industrial automation
Control	ZigBee will make possible flexible management of lighting, heating and cooling systems from anywhere in the home. It will automate control of multiple home systems to improve conservation, convenience and safety.	ZigBee will integrate and centralise the management of lighting, heating, cooling and security. It will automate the control of multiple systems to improve conservation, flexibility and security.	ZigBee will extend existing manufacturing and process control systems' reliability and improve asset management by continuously monitoring critical equipment.
Conservation	ZigBee will allow capturing of highly detailed electric, water and gas utility usage data and optimise consumption of natural resources.	It will reduce the energy expenses through optimised HVAC management and allocate utility costs equitably based on actual consumption.	It will reduce the energy costs through optimised manufacturing processes and identify inefficient operation or poorly performing equipment.
Convenience/flexibility/efficiency	It will be possible to install, upgrade and network home control system without wires and to configure and run multiple systems from a single remote control.	It will be possible to reconfigure lighting systems quickly to create adaptable workspaces, extend and upgrade building infrastructure with minimal effort.	ZigBee technology will automate data acquisition from remote sensors to reduce user intervention and provide detailed data to improve preventive maintenance programmes.
Safety	Wireless sensors could be easily installed to monitor a wide variety of conditions and give automatic notification upon detection of unusual events.	With ZigBee, networking and integration of data could be done from multiple access control points. Wireless monitoring networks could be deployed to enhance perimeter protection.	It will make possible deployment of monitoring networks to enhance employee and public safety. It will also streamline data collection for improved compliance reporting.

devices that are within its range.

What's in store?

ZigBee has a very promising future ahead. Research claims that fuelled by rapid rise in home networking, ZigBee would entirely change the wireless world. It is set to squeeze this world into a single touch control.

Low-power wireless sensing and control markets, enabled by ZigBee, are experiencing exponential growth. ZigBee chipset suppliers are today shipping tens of millions of units and the market is poised to grow to hundreds of millions of units within the next few years.

ON World Inc., a provider of smart-class business intelligence on smart technology markets, in its research on more than 600 companies has found that the ZigBee market has crossed the chasm and is now

a mainstream technology. Despite a challenging economy, sales of 802.15.4/ZigBee units have increased on an average of 62 per cent a year since 2007. Moreover, chipsets using ZigBee specifications made up 75 per cent of the 802.15.4 units shipped in 2009.

ON World has identified over 350 companies that offer or are developing ZigBee products with combined annual revenues of more than \$1 trillion. ZigBee supporters include some of the world's largest companies including AT&T, GE, Hitachi, Kroger, Panasonic, Philips, Samsung, Siemens and Sony. Demonstrated by new ZigBee component suppliers such as BlueRadios, CEL, Dust Networks and SyChip, momentum is growing.

Boston-based Ember Corporation reached a key milestone in its quest to become the world's foremost provider

ZigBee has become the chosen low-power wireless standard for advanced metering and home-area networks. Consequently, ZigBee chipset shipments are projected to reach as much as 192 million units in 2014

of ZigBee technology for smart meters and home-area network products when it became the first company to ship more than ten million ZigBee wireless chips. This milestone is reflected in the company's record revenues and sales bookings, making Ember well positioned for almost 300 per cent annual revenue growth in 2010. The company develops wireless mesh networking technology (chips, software and tools) for smart energy, connected homes, and many other monitoring and control applications, enabling greener living and work environments.

Across the world, billions are being invested in smart grid programmes as part of grid modernisation and carbon emission control measures. ZigBee has become the chosen low-power wireless standard for advanced metering and home-area networks. Consequently, ZigBee chipset shipments are projected to reach as much as 192 million units in 2014, up from just over five million units in 2008, according to research firm West Technology Research Solutions.

ZigBee and the underlying IEEE 802.15.4 standard promise a low-cost, low-power and reliable wireless network technology for a wide range of control and monitoring applications within the private sphere and industrial environment. The IEEE 802.15.4 PHY has been specified with focus on highly integrated and low-power chip solutions which promote simple and flexible end-application designs at low cost. ●